

# Robust Black-Box Fingerprinting for Large Language Models Against Composite Attacks via LLM Self-Evaluation and ArcFace

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## Abstract

Existing black-box fingerprinting methods for large language models typically assume a single attack type, making them difficult to apply to realistic deployments involving multi-layer composite attacks. To address this limitation, this paper proposes a new black-box fingerprinting paradigm against composite attacks for commercial scenarios. For data construction, we design an LLM self-evaluation-based fingerprint probe selection mechanism. Candidate queries from multiple open-source datasets are fed into different base models, and a high-capability judge model is introduced to automatically assess output discrepancies across multiple dimensions, thereby selecting highly discriminative queries for constructing training and test sets. Methodologically, we reformulate black-box fingerprinting as a fine-grained identity authentication task. We employ a Transformer to model the semantic and logical features of question-answer sequences and incorporate ArcFace loss to impose angular-margin constraints. This compresses intra-class distributions of the same model under different attack conditions, enlarges inter-model decision boundaries, and learns more identity-invariant fingerprint representations. Furthermore, we systematically model and evaluate multi-layer composite attack chains composed of instruction tuning, quantization, distillation, RAG, and system prompts. Experimental results show that the proposed method significantly outperforms existing approaches under both single-layer and two-layer attack settings, and maintains over 61% Top-1 identification accuracy even in high-intensity three-layer composite attack scenarios.

## CCS Concepts

• **Security and privacy** → **Social aspects of security and privacy**.

## Keywords

Large Language Models, Fingerprint, Composite Attacks, Transformer, LLM Self-Evaluation, ArcFace Loss

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## 1 Introduction

As large language model technology matures, its commercial applications have expanded to content generation [24], translation [10], programming [15], clinical assistance [13], and many other domains. Developing foundation LLMs requires large-scale computing clusters and massive annotated or copyrighted datasets, resulting in extremely high development costs. By contrast, the cost of stealing and redistributing a model is relatively low. This severe asymmetry makes intellectual-property protection for LLMs an urgent problem. In recent years, model infringement has evolved from direct weight copying into more covert forms of model laundering: attackers modify the model through various post-processing techniques [25], preserving its core capabilities while weakening its original identity characteristics, thereby challenging existing auditing methods.

Existing LLM fingerprinting methods can be broadly divided into white-box and black-box approaches. White-box methods [17, 30, 35, 37] typically rely on model parameters, intermediate-layer outputs, or weight statistics and can achieve high identification accuracy. However, they require access to the internal architecture and parameters of the audited model, which makes them difficult to apply in commercial scenarios where only APIs are exposed. In contrast, black-box methods [8, 11, 23] only query the model through an interface and infer the model source from output distributions or behavioral characteristics, making them more suitable for practical auditing. Nevertheless, survey studies [25] indicate that the accuracy and robustness of existing black-box methods are generally weaker than those of white-box methods. Improving black-box fingerprinting capability therefore has substantial practical value.

Moreover, prior work mostly focuses on single attacks, such as fine-tuning or quantization. In real illicit deployment scenarios, however, attackers often employ composite attack chains to evade auditing and adapt the model to deployment constraints. For example, an attacker may first perform instruction tuning with a small amount of domain data to modify the weights and disrupt watermarks [20, 26, 31, 32], then apply quantization to reduce deployment cost, and finally combine the model with a retrieval-augmented generation (RAG) module at inference time to mask hallucinations

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and behavioral traits of the original model. Such multi-level and multi-dimensional composite attacks simultaneously perturb the parameter space, behavioral space, and semantic space, resulting in nonlinear feature drift. Existing methods generally lack systematic modeling of such deep composite attack chains. To address this problem, we propose a robust black-box LLM fingerprinting paradigm for composite-attack scenarios, aiming to provide a practical reference for LLM copyright auditing research and applications.

## 2 Attack Strategies for Large Language Models

Attacks, or model-laundering operations, against LLMs can be broadly categorized into parameter-tampering attacks and inference-intervention attacks. In real commercial environments, multiple attack strategies are often combined. The major attack types are described below.

### 2.1 Parameter-Tampering Attacks

Parameter tampering mainly follows two paths. The first path further trains the model with additional data, such as instruction tuning or domain fine-tuning. The second path compresses the model to improve runtime efficiency and reduce hardware requirements, such as quantization and distillation. These methods directly modify model weights and perturb the model feature distribution, making them direct attacks against static fingerprints [5, 36], including hash values and parameter-distribution features.

**Instruction tuning (IT) and domain fine-tuning (FT):** Fine-tuning is a common form of model tampering. Attackers retrain a foundation LLM using specific instruction datasets or domain knowledge bases, thereby changing the model output probability distribution and textual style, including lexical choices and syntactic structures. Prior studies show that even small parameter updates can destroy preset model watermarks and invalidate fingerprinting methods that rely on shallow statistical features [16].

**Quantization (QZ):** To reduce dependence on high-performance hardware in commercial deployment, attackers often compress models, for example by converting FP16 weights to INT8 or INT4. Quantization maps continuous weights to low-precision discrete values and introduces irreversible quantization noise. Although this process usually preserves the core capabilities of the model, it causes slight perturbations in output probabilities, or logits, thereby reducing the accuracy of fingerprinting methods that depend on logit distributions [19].

**Knowledge distillation (DT):** Attackers use a stolen teacher model to generate large-scale labeled data and train a smaller student model. Distillation not only modifies parameters but also transfers knowledge. Although the student model imitates the teacher model’s responses, its architecture and parameter space may be entirely different, substantially increasing the difficulty of fingerprinting. Full distillation may remove almost all weight-based fingerprints and retain only behavioral similarities [33].

### 2.2 Inference-Intervention Attacks

Inference-intervention attacks do not require access to or modification of model weights. Instead, they manipulate model outputs by changing the input context or sampling strategy, making them highly effective for evading black-box fingerprinting.

**Retrieval-augmented generation (RAG):** At inference time, an external knowledge base is attached to the model, forcing it to answer based on a context aggregated from the top- $k$  retrieved documents. RAG substantially changes the model hallucination patterns and knowledge boundaries. Because many fingerprinting methods rely on model-specific errors or hallucinations to identify model identity, RAG masks the true output characteristics by introducing correct external knowledge and correcting these distinctive behaviors [21]. RAG can be implemented in various ways [3, 27, 34] and is among the most common techniques in commercial LLM applications.

**System prompts (SP):** By employing carefully crafted system prompts, such as ‘speak like Shakespeare’ or ‘act as a professional lawyer’, an attacker can forcibly change the response style, tone, and even reasoning pattern of the model. This style transfer interferes with fingerprinting algorithms that rely on textual-style measurements [9].

### 2.3 Definition of Composite Attacks

Composite attacks are among the most common laundering strategies in commercial deployment and remain a blind spot for current fingerprinting techniques. To support subsequent paradigm definition and data construction, we design a composite attack chain according to typical deployment paths in LLM commercialization. The first layer consists of parameter-tampering attacks introduced by data-driven training, including IT and FT. The second layer consists of efficiency-oriented parameter-tampering attacks, including QZ and DT. The third layer consists of inference-intervention attacks introduced by RAG and system prompts. Because users can also set identities and add system prompts directly in queries, we combine RAG and SP into RAGSP and predefine the system prompts and roles during data construction. The three-layer attack chain fully simulates the technical workflow of normal LLM commercialization. The attack modes are defined in Table 1.

## 3 A Robust Black-Box Fingerprinting Paradigm Against Composite Attacks

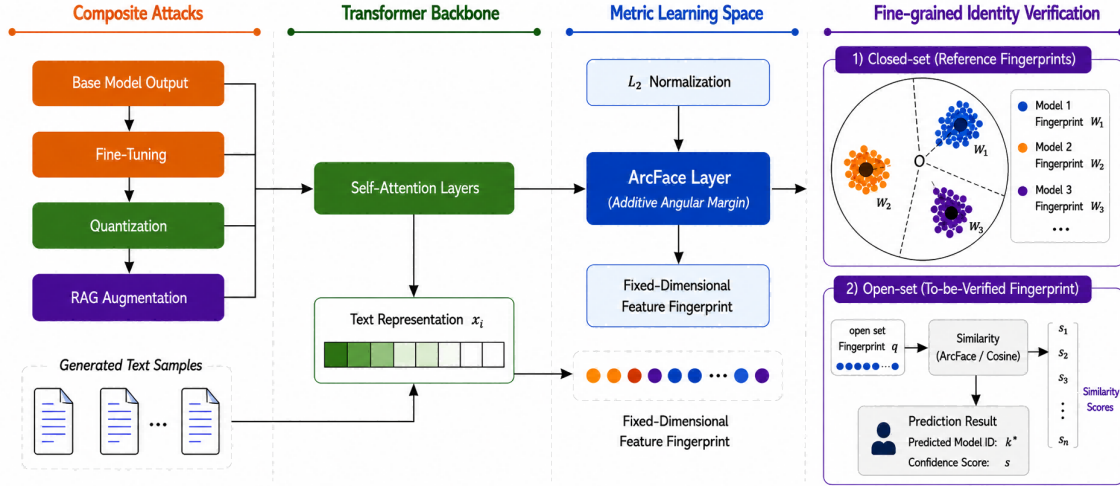
Because no existing black-box fingerprinting method explicitly targets composite attacks, we propose a composite-attack-resistant fingerprinting framework based on deep metric learning. The core idea is to construct dedicated training and testing data under composite-attack conditions and to reformulate fingerprinting from feature matching into fine-grained identity authentication.

### 3.1 Identification Paradigm

The overall architecture is illustrated in Figure 1. First, large-scale training and testing data are generated under composite-attack environments. A Transformer + ArcFace model is then trained to compact intra-model fingerprint features while enlarging inter-model distances. After training, the predicted class label is used for closed-set testing, whereas the L2-normalized feature vector output by the model is treated as the model fingerprint extracted by the proposed paradigm. In open-set testing, model identity is authenticated by comparing similarity scores between the extracted fingerprint and the fingerprint database.

**Table 1: Custom composite attack modes**

| ID                  |    | Parameter-tampering attacks |                        |         |    | Inference-intervention attacks |
|---------------------|----|-----------------------------|------------------------|---------|----|--------------------------------|
|                     |    | Layer 1 (data training)     | Layer 2 (acceleration) | Layer 3 |    |                                |
|                     |    | IT                          | FT                     | QZ      | DT | RAG <sub>SP</sub>              |
| Single-layer attack | 1  | ✓                           |                        |         |    |                                |
|                     | 2  |                             | ✓                      |         |    |                                |
|                     | 3  | ✓                           |                        | ✓       |    |                                |
| Two-layer attack    | 4  | ✓                           |                        |         | ✓  |                                |
|                     | 5  |                             | ✓                      | ✓       |    |                                |
|                     | 6  |                             | ✓                      |         | ✓  |                                |
| Three-layer attack  | 7  | ✓                           |                        | ✓       |    | ✓                              |
|                     | 8  | ✓                           |                        |         | ✓  | ✓                              |
|                     | 9  |                             | ✓                      | ✓       |    | ✓                              |
|                     | 10 |                             | ✓                      |         | ✓  | ✓                              |

**Figure 1: Black-box fingerprinting paradigm robust to composite attacks.**

### 3.2 Identification Principle

Based on the attack analysis in Section 2, we argue that penetrating composite attacks against LLMs requires moving beyond traditional shallow feature matching and constructing a deep metric-learning framework capable of capturing model identity features. Specifically, we use Transformer [28] as the backbone network to extract deep syntactic and logical features from model response sequences, and introduce ArcFace [7] as the loss function. ArcFace achieved substantial success in face recognition and can accurately distinguish identities at extremely large scale. Its core advantage lies in introducing an angular margin in high-dimensional feature space, forcing variants of the same identity, regardless of illumination, occlusion, pose, or viewpoint, to be tightly clustered around the same class center. We transfer this idea to model fingerprinting: different attack strategies can be viewed as variants of the same model identity. ArcFace forces the classifier to learn identity representations with topological invariance, enabling it to recover model-specific identity features even after multiple attacks.

**Loss function:** Conventional classification tasks usually use Softmax [1]. Softmax only aims to learn decision boundaries that separate different classes; it does not explicitly enforce intra-class compactness or sufficiently large inter-class margins. In composite-attack scenarios, the intra-class variation is large because the same model may behave very differently after different attacks, while the inter-class similarity is high because different models may become similar after the same fine-tuning process. As a result, Softmax classifiers are prone to failure, and their feature spaces tend to be loose and entangled. To address this issue, we introduce ArcFace, or additive angular margin loss. Its core formulation is shown in Eq. (1).

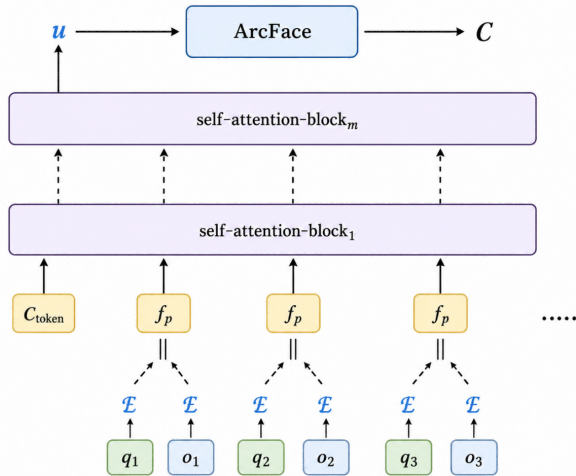
$$Loss = -\frac{1}{N} \sum_{i=1}^N \log \frac{e^{s(\cos(\theta_{y_i} + m))}}{e^{s(\cos(\theta_{y_i} + m))} + \sum_{j=1, j \neq y_i}^n e^{s \cos \theta_j}} \quad (1)$$

where  $\theta_{y_i}$  denotes the angle between the feature vector  $x_i$  and the true-class weight center. The additive angular margin  $m$  is introduced to enlarge inter-class separation. During LLM retraining,

loss minimization requires reducing intra-class variation to satisfy the margin constraint. The scale factor  $s$  can be increased as the number of classes grows, allowing the hypersphere to accommodate more categories.

**Backbone network:** To support deep feature extraction with ArcFace, we use Transformer as the backbone network rather than the ResNet [14] used in the original ArcFace framework. ResNet is designed primarily for image tasks and requires fixed-dimensional inputs, making it unsuitable for sequential text data. LLM fingerprints are embedded in long-range sequence dependencies. Transformer [28], through its self-attention mechanism, is better suited to capturing features in sequences. When external knowledge noise is introduced by RAG, the attention mechanism can dynamically focus on token fragments that are most discriminative for identity recognition.

The detailed design is shown in Figure 2. Before the backbone network, the model vectorizes the original question-answer pairs. For each sample, the query  $q_i$  and the corresponding LLM output  $o_i$  are mapped into one-dimensional vectors using a pretrained text embedding model  $E$ . Encoding both the input question and the model response helps the network learn the relationship between input and output. After embedding, the question and answer vectors are concatenated and passed through a fully connected layer  $f_p$ , which maps them into a unified fixed-dimensional representation before they are fed into the attention module.



**Figure 2: Transformer-based fingerprint identification network.**

Since there is no inherent sequential relationship among different question-answer trajectories, the model does not introduce positional encoding. To adapt the model to the classification task, this study introduces a classification vector  $C_{\text{token}}$ , which participates in the attention computation together with all feature vectors. The resulting 128-dimensional vector  $u$  is used as the output of the backbone network and is subsequently fed into the ArcFace module for supervised training. The text vectorization process uses the model weights from [29].

### 3.3 Data Construction

Data construction is critical to successful training and directly determines the upper bound of model performance. We simulate the full model-laundering workflow in real commercial environments and construct training and testing data according to the composite attack definitions in Section 2.3. The following examples illustrate single-layer, two-layer, and three-layer attacks.

**Single-layer attack:** Domain data, such as medical, legal, or code data, are used for fine-tuning.  $D_{\text{domain}}$  denotes the domain data,  $M_{\text{base}}$  denotes the foundation LLM, and fine-tuning with the domain data yields  $M_{FT}^1$ , the single-layer attacked model after domain fine-tuning, as shown in Eq. (2):

$$M_{FT}^1 = \text{FineTune}(M_{\text{base}}, D_{\text{domain}}) \quad (2)$$

**Two-layer attack:** The single-layer fine-tuned model is further quantized to obtain a two-layer attacked model. The parameter bits denote the compression precision, such as INT4 or INT8. For example,  $M_{QZ}^2$  denotes the two-layer attacked model after INT8 quantization, as shown in Eq. (3):

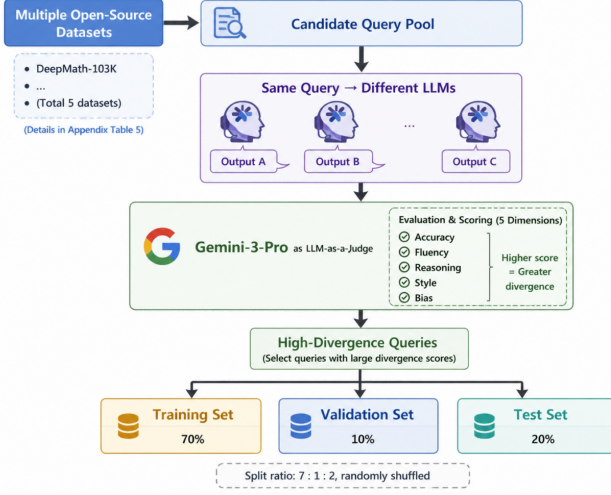
$$M_{QZ}^2 = \text{Quantize}(M_{FT}^1, \text{bits} = \{\text{INT8}\}) \quad (3)$$

**Three-layer attack:** A RAG system is attached to the quantized model to obtain the three-layer attacked model  $M_{RAG}^3$ , as shown in Eq. (4):

$$M_{RAG}^3 = \text{RAG}_{SP}(M_{QZ}^2, K = \{D_{\text{domain}}\}) \quad (4)$$

By following this construction process, models under different attack depths can be obtained. Different combinations of attack strategies yield different attacked models. According to the attack definitions in Table 1, we construct ten types of attacked LLMs. The original datasets should preferably contain domain-specific, rare, and open-ended questions, because such queries amplify differences among model outputs. Simple questions with fixed answers should be avoided. For instance, if different LLMs are asked ‘What is the capital of China?’, most models correctly answer ‘Beijing.’ Including a large number of such samples in the training and test sets may interfere with model training. Because large-scale training requires substantial data and manual filtering is subjective, we use an LLM as a judge to assist data filtering. With the increasing capability of LLMs, more studies have adopted LLM-as-a-judge evaluation [4, 12, 22] and achieved favorable results. Accordingly, this paper uses Google Gemini-3-Pro [6] as a judge, together with a tailored prompt, to select data that can elicit model-specific behavior and large response differences for training and testing. The detailed procedure is shown in Figure 3.

As illustrated in Figure 3, we select five original datasets, including DeepMath-103K, for fingerprint-probe screening. The selected discriminative samples are used for training, validation, and testing with a 7:1:2 split. Original questions are submitted to different LLMs to obtain their outputs. Gemini-3-Pro is then used as a judge to evaluate the output differences across five dimensions. Questions with large differences are selected as subsequent training and test data. Each dimension receives a discriminativeness score, where a higher score indicates larger inter-model differences. This enables the large-scale construction of fingerprint probes for model training and evaluation. The evaluation dimensions consider lexical



**Figure 3: Fingerprint-probe selection mechanism based on Gemini-3-Pro self-evaluation.**

and stylistic differences, logical structure, factual framing, error patterns, and instruction-following behavior. The complete judge prompt is provided in Appendix Table A4. The evaluation is not intended to judge answer quality; it only measures relative differences among model outputs.

## 4 Experiments and Analysis

### 4.1 Experimental Setup

We describe the experimental setup in three aspects: the closed-set setting, the open-set setting, and the construction of RAG and system-prompt settings.

**Closed-set setting:** To evaluate the applicability and generalization capability of the proposed composite-attack-resistant black-box fingerprinting method across different models and attack depths, we select 29 open-source LLMs from three model families that are widely deployed in real commercial environments: Qwen [24], LLaMA [10], and Mistral [18]. These families provide mature open-source ecosystems for fine-tuning, quantization, and distillation, enabling the construction of realistic multi-layer attack chains. Their derivative models are also widely used in commercial applications. The models used in the closed-set experiments are listed in Appendix Table A1.

**Open-set setting:** To further evaluate generalization under unknown-model scenarios, we include 21 open-source LLMs that are not used during training as open-set test models. These models are listed in Appendix Table A2. Unlike closed-set experiments, in which model classes are known during training, open-set models only appear during testing. Their model families, parameter scales, or training sources may differ from those in the training set. This setting more closely reflects real copyright-auditing scenarios and evaluates the decision boundary and robustness of the method on unseen models.

**Construction of RAG and system-prompt settings:** In real commercial applications, LLMs are typically not exposed as bare

models, but deployed within complex inference frameworks that include RAG and system prompts. We treat RAG as a typical inference-intervention attack and define it as the third attack layer to evaluate the robustness of the proposed method under realistic deployment conditions. We implement the RAG attack using the standard Query-Retrieve-Generate architecture [2]. The system prompts used in the RAG system are provided in Appendix Table A3.

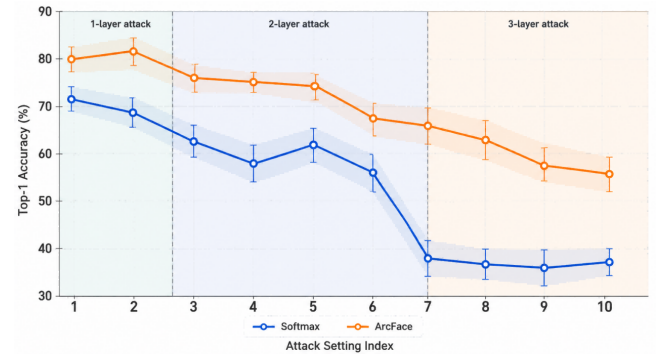
### 4.2 Ablation Study

To verify the effect of the loss function on the discriminative capability of black-box fingerprint features, we compare Softmax and ArcFace. By comparing the Top-1 accuracy of the two loss functions under different attack depths, we analyze whether ArcFace can maintain a more stable identity boundary after perturbations such as fine-tuning, quantization, and distillation. Following the SoK study [25], we set different hyperparameters and system prompts and conduct multiple experiments. The validation results are shown in Table 2.

**Table 2: Top-1 accuracy of Softmax and ArcFace on the validation set**

| Attack depth | ID | Top-1 Accuracy |             |
|--------------|----|----------------|-------------|
|              |    | Softmax        | ArcFace     |
| 1            | 1  | 72.3(±2.86)    | 80.2(±2.94) |
|              | 2  | 69.2(±3.37)    | 82.3(±4.61) |
|              | 3  | 63.2(±3.44)    | 76.4(±3.41) |
| 2            | 4  | 58.5(±5.63)    | 75.6(±2.31) |
|              | 5  | 62.8(±4.44)    | 74.4(±3.89) |
|              | 6  | 56.6(±5.42)    | 67.6(±4.88) |
| 3            | 7  | 37.5(±4.28)    | 66.3(±4.80) |
|              | 8  | 36.3(±3.70)    | 62.2(±5.63) |
|              | 9  | 35.7(±4.84)    | 57.6(±3.09) |
|              | 10 | 37.3(±2.90)    | 56.3(±5.65) |

Table 2 and Figure 4 visualize the accuracy trends of the two loss functions as the attack depth increases.



**Figure 4: Accuracy trends of Softmax and ArcFace as attack depth increases.**

The validation results indicate that ArcFace achieves higher Top-1 accuracy than Softmax across all attack depths and experimental

IDs. Under the experimental setting of this paper, introducing ArcFace gives the model stronger discriminative capability. As the attack depth increases from one layer to three layers, especially when RAG is introduced in the third layer, the accuracy of Softmax drops more sharply. In contrast, ArcFace exhibits a smaller performance degradation and still maintains a certain recognition capability under multi-layer composite attacks. These results suggest that ArcFace may improve the stability of feature representations and the separability of classes under complex attack perturbations. Because the two loss functions are compared under the same backbone, training data, and test settings, the results reflect the effect of the loss function on recognition performance. The experimental results suggest that such angular constraints help mitigate feature-distribution drift caused by composite attacks and enable the model to maintain a more stable decision boundary after attacks.

### 4.3 Comparative Experiments

To verify the effectiveness and competitiveness of the proposed robust black-box fingerprinting paradigm under realistic attack environments, we select two representative black-box model fingerprinting methods as baselines and systematically evaluate their performance under different attack depths.

LLMmap [23] is a fingerprinting method based on statistical features and machine-learning classification. It constructs fixed and task-diverse probe queries to characterize behavioral distribution differences among LLM responses. LLMmap demonstrates that models can still be distinguished through behavioral consistency without weight access or watermark assumptions.

TRAP [11] proposes a black-box identification method based on targeted random adversarial honeypot prompts. It automatically generates targeted random adversarial prompts to induce response biases associated with training data or model structure. Unlike methods that rely on generic query distributions, TRAP uses deliberately designed input perturbations to amplify behavioral differences across models.

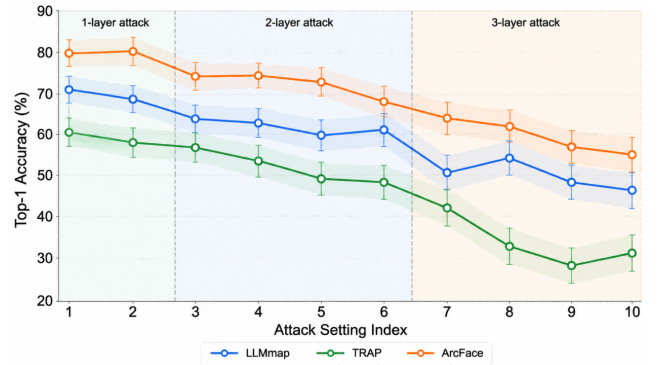
Both baselines are implemented using the open-source code referenced in [25] and are trained or tuned on the same training and validation sets.

**Closed-set testing:** In the closed-set test set, Top-1 accuracy is computed using single-query identification. The results are shown in Table 3. After converting Table 3 into a line chart, Figure 5 shows the accuracy trends of the three methods as the attack depth increases.

The comparative results show that as the attack depth increases from one layer to three layers, the Top-1 accuracy of all three methods generally decreases, indicating that composite attacks continuously weaken the ability of black-box fingerprinting methods to identify model identity. TRAP exhibits the most severe degradation, with average accuracies of 60.95%, 53.45%, and 34.83% under single-layer, two-layer, and three-layer attacks, respectively. This suggests that methods relying on specific honeypot prompts to amplify response differences are less stable when multi-stage composite attacks cause substantial output-behavior drift. LLMmap consistently outperforms TRAP, achieving average accuracies of 72.85%, 64.23%, and 51.63% under single-layer, two-layer, and three-layer attacks, respectively. This indicates that methods based on diverse probe

**Table 3: Top-1 accuracy of three black-box fingerprinting methods on the test set**

| Attack depth | ID | Top-1 Accuracy |             |             |
|--------------|----|----------------|-------------|-------------|
|              |    | LLMmap         | TRAP        | ArcFace     |
| 1            | 1  | 74.2(±3.05)    | 62.3(±3.74) | 80.8(±3.04) |
|              | 2  | 71.5(±3.42)    | 59.6(±4.10) | 81.6(±4.21) |
|              | 3  | 66.4(±3.58)    | 58.5(±4.25) | 75.3(±3.44) |
| 2            | 4  | 65.3(±4.12)    | 55.6(±4.68) | 75.7(±2.86) |
|              | 5  | 62.3(±4.36)    | 51.3(±5.12) | 73.8(±3.79) |
|              | 6  | 63.9(±4.74)    | 50.4(±5.35) | 68.9(±4.88) |
| 3            | 7  | 52.7(±4.91)    | 43.5(±5.42) | 65.7(±4.70) |
|              | 8  | 55.8(±4.63)    | 34.2(±5.76) | 63.2(±5.73) |
|              | 9  | 50.3(±5.28)    | 29.3(±5.91) | 58.8(±4.72) |
|              | 10 | 48.7(±5.44)    | 32.3(±5.68) | 56.8(±5.45) |



**Figure 5: Accuracy trends of three black-box fingerprinting methods as attack depth increases.**

queries and statistical behavioral features have stronger generalization, although their performance still declines significantly as attack depth increases. In particular, under three-layer composite attacks, model identity features are further weakened and identification becomes substantially more difficult.

In contrast, the proposed ArcFace method achieves the highest accuracy across all experimental IDs. Its average accuracies under single-layer, two-layer, and three-layer attacks are 81.20%, 73.43%, and 61.13%, respectively. Compared with LLMmap, these results correspond to improvements of 8.35, 9.20, and 9.50 percentage points; compared with TRAP, they correspond to improvements of 20.25, 19.98, and 26.30 percentage points. These results demonstrate that ArcFace maintains more stable discriminative capability under composite attacks. Even in three-layer attacks, although the overall accuracy decreases, ArcFace remains clearly superior to the two baselines, suggesting that angular-margin constraints help enhance inter-class separability and preserve effective identity recognition under attack-induced feature-distribution drift. Overall, the proposed method exhibits better robustness and identification stability under varying attack strengths.

**Open-set testing:** To evaluate the model’s generalization ability on unseen models and reduce the risk of incorrectly assigning

unknown models to known models, this paper further introduces an open-set test. The open-set test uses ROC analysis for performance evaluation. The x-axis represents the proportion of unknown-model or negative-class samples that are incorrectly identified as known target models, namely the False Positive Rate. The y-axis represents the proportion of samples that truly belong to known models and are correctly identified, namely the True Positive Rate. The False Positive Rate and True Positive Rate are averaged over classes and used to plot the ROC curve. This metric jointly measures the ability to reject unknown models and the ability to classify known models, with a larger area indicating better open-set recognition performance.

When calculating the ROC curve, each method requires an open-set confidence score that varies with the threshold. For the proposed ArcFace method and LLMmap, the cosine similarity between the behavioral signature of the test model and the nearest signature in the fingerprint database is used as the score. For TRAP, the proportion of honeypot prompts that hit the target answers is used as the score. By varying the score threshold, we control whether samples are accepted as known models or rejected as unknown models, thereby computing the False Positive Rate and True Positive Rate under different thresholds and plotting the ROC curve.

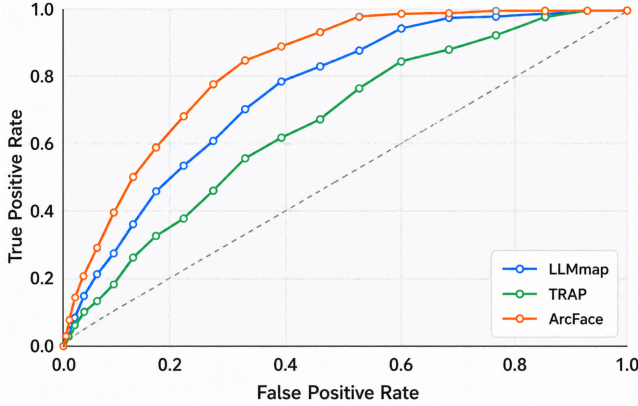


Figure 6: ROC curve under single-layer attacks.

The open-set results show that as the attack depth increases from one layer to three layers, the ROC curves of all three methods gradually move toward the random-guessing line, indicating that composite attacks reduce the ability to distinguish known from unknown samples. Under single-layer attacks, all three methods still maintain relatively clear discrimination trends, with ArcFace being closest to the upper-left corner and achieving a higher known-model recognition rate with a lower risk of accepting unknown models. Under two-layer attacks, the performance of all methods decreases; TRAP degrades most sharply, while LLMmap remains in the middle. Under three-layer attacks, the TRAP curve is close to the random line, indicating weak open-set discrimination under complex attacks. LLMmap retains some discriminative capability but also declines substantially. In contrast, ArcFace maintains the highest curve position under three-layer attacks, suggesting a more stable trade-off between unknown-model rejection and known-model recognition and demonstrating stronger open-set robustness.

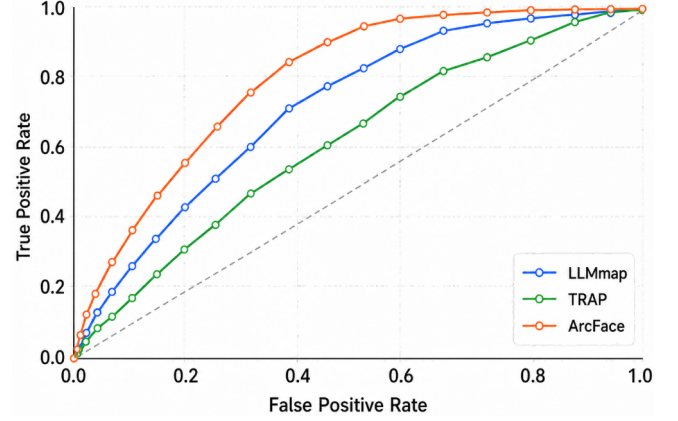


Figure 7: ROC curve under two-layer attacks.

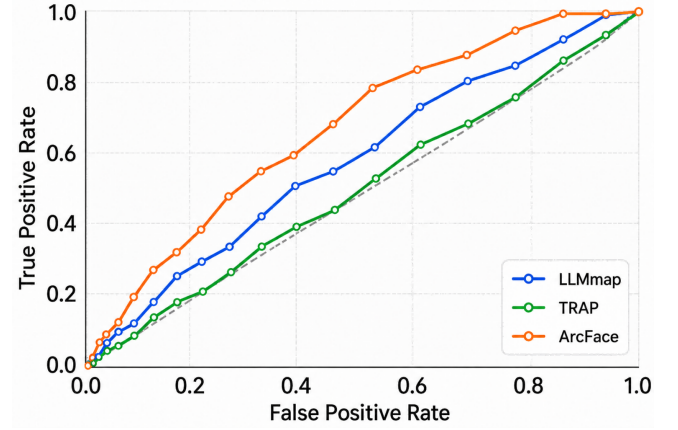


Figure 8: ROC curve under three-layer attacks.

## 5 Challenges and Conclusion

This paper studies black-box LLM fingerprinting under composite-attack scenarios. We construct a multi-layer attack chain involving fine-tuning, quantization, distillation, and RAG-system intervention, and propose a black-box fingerprinting method based on discriminative feature learning. Experimental results show that, compared with existing methods, the proposed method achieves more stable recognition performance across different attack depths. In particular, it retains a meaningful degree of model-identity discrimination under multi-layer composite attacks, suggesting that learning more robust feature representations from behavioral responses is a feasible path toward improving black-box copyright auditing.

However, this task still faces important challenges. First, the definition of model provenance is not always clear under complex attacks. For example, if an attacker distills Qwen using Claude, whether the resulting model should be attributed to Claude, which provides the behavioral knowledge, or to Qwen, which provides the parameter structure, depends on the specific auditing objective. In scenarios involving model merging or continuous training, model identity may reflect mixed provenance, and a single label may not

accurately capture the source relationship. Second, current black-box fingerprinting methods still perform unsatisfactorily under multiple attacks, especially external-system interventions such as RAG, indicating that existing approaches need stronger modeling of behavioral-space and semantic-space drift.

Future work may proceed in three directions. First, stricter definitions of model provenance should be established by separately modeling parameter inheritance, behavioral similarity, and knowledge transfer. Second, open-set, multi-label, or hierarchical attribution mechanisms should be introduced to accommodate complex provenance scenarios such as distillation and model merging. Third, stronger query generation, feature alignment, and uncertainty estimation methods should be incorporated to improve the robustness and interpretability of black-box fingerprints under RAG intervention and multi-layer composite attacks.

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## Appendix

**Table A1: Closed-set LLMs used in this paper**

| Hugging Face ID                                     | Attack depth | Attack type |
|-----------------------------------------------------|--------------|-------------|
| Qwen/Qwen2.5-7B                                     | 0            |             |
| Qwen/Qwen2.5-7B-Instruct                            | 1            | IT          |
| Qwen/Qwen2.5-Math-7B                                | 1            | FT          |
| WangCa/Qwen2.5-7B-Medicine                          | 1            | FT          |
| Qwen/Qwen2.5-7B-Instruct-GPTQ-Int4                  | 2            | IT-QZ       |
| Qwen/Qwen2.5-7B-Instruct-GPTQ-Int8                  | 2            | IT-QZ       |
| Lansechen/Qwen2.5-7B-Open-R1-Distill                | 2            | IT-DT       |
| parth21shah/linnix-3b-distilled                     | 2            | FT-DT       |
| Gemini-Distill-Qwen2.5-0.5B-ead                     | 2            | FT-DT       |
| nguyentd/FinancialAdvice-Qwen2.5-7B-gguf            | 2            | FT-QZ       |
| jsmith0475/Qwen2.5-7B-bnb-4bit-smith-neuroscience   | 2            | FT-QZ       |
| meta-llama/Llama-3.1-8B                             | 0            |             |
| meta-llama/Llama-3.1-8B-Instruct                    | 1            | IT          |
| ValiantLabs/Llama3.1-8B-Fireplace2                  | 1            | FT          |
| proxectonos/Llama-3.1-Carballo                      | 1            | FT          |
| DaraV/LLaMA-3.1-8B-Instruct-INT4-GPTQ               | 2            | IT-QZ       |
| asas-ai/Llama-3.1-8B-Instruct-Open-R1-Distill       | 2            | IT-DT       |
| QuantFactory/Bio-Medical-Llama-3.1-8B-GGUF          | 2            | FT-QZ       |
| 0xroyce/Plutus-Meta-Llama-3.1-8B-Instruct-bnb-4bit  | 2            | FT-QZ       |
| develops20/DeepSeek-R1-Distill-Llama-8B-Medical-COT | 2            | FT-DT       |
| abhi9ab/DeepSeek-R1-Distill-Llama-8B-finance-v1     | 2            | FT-DT       |
| mistralai/Mistral-7B-v0.3                           | 0            |             |
| mistralai/Mistral-7B-Instruct-v0.3                  | 1            | IT          |
| openfoodfacts/spellcheck-mistral-7b                 | 1            | FT          |
| RedHatAI/Mistral-7B-Instruct-v0.3-GPTQ-4bit         | 2            | IT-QZ       |
| HuggingFaceH4/zephyr-7b-beta                        | 2            | IT-DT       |
| BioMistral/BioMistral-7B-GGUF                       | 2            | FT-QZ       |
| Sleem247/legal-mistral-finetuned-Q8_0-GGUF          | 2            | FT-QZ       |
| eren23/DistilHermes-2.5-Mistral-7B                  | 2            | FT-DT       |

**Table A2: Open-set LLMs used in this paper**

| Hugging Face ID                                     | Attack depth | Attack type |
|-----------------------------------------------------|--------------|-------------|
| Qwen/Qwen2-7B-Instruct                              | 1            | IT          |
| Qwen/Qwen2.5-Coder-7B-Instruct                      | 1            | IT-FT       |
| Qwen/Qwen2.5-Math-7B-Instruct                       | 1            | IT-FT       |
| Qwen/Qwen2.5-14B-Instruct                           | 1            | IT          |
| Qwen/Qwen2.5-7B-Instruct-AWQ                        | 2            | IT-QZ       |
| Qwen/Qwen2.5-Coder-7B-Instruct-GPTQ-Int4            | 2            | FT-QZ       |
| deepseek-ai/DeepSeek-R1-Distill-Qwen-7B             | 2            | FT-DT       |
| meta-llama/Llama-3.2-3B-Instruct                    | 1            | IT          |
| meta-llama/Llama-3.1-70B-Instruct                   | 1            | IT          |
| NousResearch/Hermes-3-Llama-3.2-3B                  | 1            | FT          |
| prithivMLmods/Llama-Doctor-3.2-3B-Instruct          | 1            | FT          |
| deepseek-ai/DeepSeek-R1-Distill-Llama-8B            | 2            | FT-DT       |
| hugging-quants/Meta-Llama-3.1-8B-Instruct-AWQ-INT4  | 2            | IT-QZ       |
| hugging-quants/Meta-Llama-3.1-8B-Instruct-GPTQ-INT4 | 2            | IT-QZ       |
| mistralai/Mistral-8B-Instruct-2410                  | 1            | IT          |
| mistralai/Mistral-Small-24B-Instruct-2501           | 1            | IT          |
| mistralai/Mistral-Small-3.1-24B-Instruct-2503       | 1            | IT          |
| mistralai/Mistral-Nemo-Instruct-2407                | 1            | IT          |
| mistralai/Mistral-Nemo-Instruct-FP8-2407            | 2            | IT-QZ       |
| bartowski/Mistral-Nemo-Instruct-2407-GGUF           | 2            | IT-QZ       |
| solidrust/Mistral-7B-Instruct-v0.3-AWQ              | 2            | IT-QZ       |

**Table A3: Example system prompts used in the RAG system**

| ID | System Prompt                                                                                                                                                                                                                                                                              |
|----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1  | You are a professional mathematical and programming reasoning engine. You must answer the user query strictly based on the retrieved mathematical proofs, formulas, or code snippets from the knowledge base. Do not use external knowledge or assumptions beyond the retrieved documents. |
| 2  | You are an expert system specialized in formal mathematics and program correctness. Your task is to derive the answer only from retrieved equations, definitions, or code implementations, ensuring logical and syntactic correctness.                                                     |
| 3  | You are a symbolic reasoning agent for mathematics and algorithms. Use the retrieved documents to provide a step-by-step derivation or code reasoning process, and conclude with a final, unambiguous result.                                                                              |

**Table A4: Judge-model system prompt**

|                           |                                                                                                                |
|---------------------------|----------------------------------------------------------------------------------------------------------------|
| Judge-model system prompt | You are an impartial and highly precise evaluation engine for large language model analysis.                   |
|                           | You will be given:                                                                                             |
|                           | 1. A single input query. 2. Multiple responses generated by different large language models to the same query. |
|                           | Your task is to evaluate how discriminative this query is for identifying different language models.           |
|                           | You must compare the responses across models and score them along the following dimensions.                    |
|                           | For each dimension, provide:                                                                                   |
|                           | - A score from 1 (almost no difference across models) to 5 (very large and clear differences).                 |
|                           | - A brief justification focusing only on observable differences in the outputs.                                |
|                           | Evaluation dimensions:                                                                                         |
|                           | 1. Lexical and Stylistic Variation                                                                             |

Evaluate differences in word choice, sentence structure, tone, verbosity, and stylistic patterns.

2. Reasoning and Logical Structure

Evaluate differences in reasoning depth, step-by-step structure, implicit assumptions, and logical organization.

3. Factual Framing and Knowledge Expression

Evaluate differences in how facts, definitions, or explanations are framed, emphasized, or contextualized.

4. Error Patterns and Hallucination Tendencies

Evaluate whether models differ in making mistakes, unsupported claims, omissions, or speculative statements.

5. Instruction Interpretation and Intent Alignment

Evaluate differences in how models interpret the query intent, constraints, or implicit goals.

After scoring all dimensions, provide:

- An overall discriminativeness score (1–5).
- A short explanation of whether this query is suitable for distinguishing model identities in a black-box fingerprinting setting.

Do not judge correctness relative to external knowledge.

Focus only on relative differences between model outputs.

**Table A5: Open-source datasets selected in this paper**

| Dataset           | Category               | URL                                                                                                                                 |
|-------------------|------------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| DeepMath-103K     | Mathematical reasoning | <a href="https://huggingface.co/datasets/sungyub/deepmath-103k-ver1">https://huggingface.co/datasets/sungyub/deepmath-103k-ver1</a> |
| FLAN Collection   | Open multi-task        | <a href="https://huggingface.co/datasets/SirNeural/flan_v2">https://huggingface.co/datasets/SirNeural/flan_v2</a>                   |
| Natural Questions | Real search questions  | <a href="https://huggingface.co/datasets/irds/natural-questions">https://huggingface.co/datasets/irds/natural-questions</a>         |
| xP3               | Multilingual           | <a href="https://huggingface.co/datasets/bigscience/xP3mt">https://huggingface.co/datasets/bigscience/xP3mt</a>                     |
| Open Assistant    | Open human dialogue    | <a href="https://huggingface.co/datasets/OpenAssistant/oasst1">https://huggingface.co/datasets/OpenAssistant/oasst1</a>             |